

Juan PINEDA-JARAMILLO, PhD

AI Engineer | LLM Engineer – Agentic Systems, RAG & Production GenAI

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AREAS OF EXPERTISE

- Agentic AI & LLM Engineering:
 - Multi-Agent Orchestration: Production multi-agent systems with LangGraph and LangChain, including an in-process agent-to-agent (A2A) protocol for cross-domain evidence exchange over shared state. Supervisor and router patterns, tool calling, agent memory, and self-correcting agent loops.
 - Prompt Engineering & Structured Output: Prompt-as-configuration (versioned YAML templates), few-shot / zero-shot design, model grounding, strict Pydantic schema enforcement, model routing and fallback. Hands-on with Azure OpenAI (GPT-5.x), OpenAI, AWS Bedrock (Claude, Titan, Nova Pro) and HuggingFace models.
 - RAG & Retrieval: GraphRAG over Neo4j knowledge graphs (text-to-Cypher retrieval), vector search with pgvector and FAISS, semantic example selection, multimodal ingestion (PPTX, PDF, DOCX, XLSX and slide-to-image vision), and source citations resolved against a catalog for evidence traceability.
 - LLM Safety & Guardrails: Defence-in-depth guardrail frameworks: prompt-injection detection (regex and semantic), PII/PHI redaction (Presidio, spaCy), data-exfiltration and system-prompt-leakage blocking, schema and citation validation, per-run cost and token ceilings, model-version pinning, and immutable hashed audit trails.
 - Evaluation & Observability: Ground-truth eval harnesses, deterministic oracles combined with LLM-as-judge, adversarial red-team corpora, offline regression suites driven by mock LLMs, tracing with Arize Phoenix, OpenTelemetry, LangSmith and Langfuse, and per-run token, latency and cost tracking.
 - Text2SQL & Neuro-Symbolic Systems: Schema-agnostic SQL agents that discover schemas at runtime and recover from their own errors, combining LLM reasoning with deterministic validation and read-only SQL enforcement.
- MLOps & LLM Delivery: Containerised FastAPI services, Kubernetes/Helm deployment, CI/CD (GitLab, Bitbucket, Jenkins, Azure DevOps), async persistence with SQLAlchemy and Alembic, structured logging, and cost/quality monitoring in production.
- Traditional Machine Learning: Classification, regression, time series analysis, clustering, dimensionality reduction, anomaly detection, mixed-integer optimization, explainability (SHAP, Sobol).
- Natural Language Processing (NLP): Sentiment analysis, text classification, topic modelling.
- Research & Development
- Programming and Development Tools:
 - Python (expert): FastAPI, Pydantic, LangChain, LangGraph, SQLAlchemy, Pandas, NumPy, Scikit-learn, Matplotlib, Kedro, pytest, structlog
 - SQL (advanced), Excel (advanced)
 - Software architecture: hexagonal (ports & adapters), domain-driven design, async services, REST APIs and microservices
- Cloud Platforms:
 - Azure: Azure OpenAI, Databricks, Azure SQL, ADF, DevOps
 - AWS: Bedrock, SageMaker, EC2, S3, Athena, EMR, RDS, Redshift
 - GCP: Vertex AI, BigQuery, Cloud Run, Cloud Storage, Cloud SQL, Dataproc, Dataflow, AutoML, Data Fusion
- Data & Platform Engineering:
 - PostgreSQL + pgvector, Neo4j, Snowflake, MySQL, PySpark, Apache Airflow, Apache Beam, MLflow
 - Docker, Kubernetes & Helm, Jenkins, GitLab CI, Bitbucket Pipelines, Azure DevOps, Git, Bash
 - Quality tooling: pytest, ruff, black, mypy, pre-commit, Alembic
- Experience with agile methodologies: Azure DevOps, Jira
- Languages: Spanish (native). English (advanced). French (limited)

PROFESSIONAL EXPERIENCE

AI Engineer | LLM Engineer (2015-Present)

AI/LLM Engineer with a PhD and 10+ years in applied ML, now focused on production agentic systems: multi-agent architectures, RAG over knowledge graphs and vector stores, and the guardrail, evaluation and observability layers that make them safe enough for regulated enterprise use. Experience across pharmaceuticals, aviation, finance, mining, logistics and retail.

Pharmaceutical & Healthcare

- **Client: Johnson & Johnson (United States of America) via ITMAGINATION (Poland)**
Building agentic AI systems for enterprise architecture governance and self-serve analytics, alongside machine learning for pharmacovigilance.
ARIA - ARB Intake Verifier: multi-agent system that automates J&J's Architecture Review Board. It ingests solution-architecture submissions (PPTX, PDF, DOCX and diagrams), reviews them against the enterprise standards library and returns a scored, fully cited report, cutting a review that took a reviewer a full working day down to roughly 10 minutes.
 - Multi-Agent Architecture: Four domain architect agents (Business, Technology, Information, Data Science & AI) running on a shared quality-agent engine, orchestrated through a LangGraph state machine, with an in-process agent-to-agent (A2A) protocol in which each peer answers once and agents revise their findings with the evidence returned. Built on LangGraph and LangChain.
 - Multimodal GraphRAG: Agent reasoning grounded in a Neo4j knowledge graph of the enterprise architecture inventory (LeanIX, text-to-Cypher retrieval) and a pgvector store over the standards library, with slide-to-image vision so agents also assess diagrams, and source-ID citations on every finding resolved against the catalog.

- LLM Safety Framework: Designed and implemented 19 production guardrails at graph-node boundaries and targeted points, non-blocking by design (warn and audit, human in the loop): prompt injection, PII/PHI redaction (Presidio), data-exfiltration and prompt-leakage detection, schema and citation validation, per-run LLM-call and token ceilings, SHA-256 hashed audit trail, among others.
- Engineering & Delivery: Async FastAPI service with PostgreSQL and Alembic, connectors for SharePoint, LeanIX and Confluence, Arize Phoenix / OpenTelemetry tracing with per-call cost tracking, a run-consistency judge, and an offline regression suite of about 590 tests driven by mock LLMs.
- Tech stack: Python, LangGraph, LangChain, Azure OpenAI (GPT-5.x), FastAPI, Pydantic, Neo4j, PostgreSQL + pgvector, SQLAlchemy, Alembic, Docker, Kubernetes, Helm

Enterprise Text2SQL Agent (proof of concept): self-serve natural-language querying of enterprise databases, so business users get answers without depending on a SQL-fluent data analyst.

- Agent Design: A single tool-calling agent that discovers the schema at runtime, writes read-only SQL and recovers from its own errors by receiving database failures back as tool results - onboarding a new database is a connection-string change, with zero schema-specific prompt engineering.
- Hexagonal Architecture & Guardrails: The agent depends only on LLMPort and DatabasePort contracts, so switching LLM vendor or database platform means writing an adapter; read-only SQL validation, step budget and row caps, plus a full offline regression suite driven by a scripted mock LLM (no API spend).
- Tech stack: Python, Azure OpenAI, LangChain, SQLAlchemy, Pydantic, structlog, pytest.

SRS Summarizer: LLM service that writes persona-specific narrative summaries (CPO, Value Chain Leader, Category Manager) of supply-disruption events for J&J's Supply Risk Sensing dashboards; proof of concept heading to the DEV environment.

- Compute first, narrate second: every figure is computed by deterministic read-only SQL into a fact sheet and the model only narrates that fact sheet; null facts are dropped before the call and out-of-scope events return a fixed message with no model call.
- Hexagonal Architecture: the service depends only on LLMPort, DatabasePort and TracePort contracts, with interchangeable OpenAI SDK and LangChain adapters and one SQL fact engine that runs on SQLite in tests and Databricks SQL in the target environment.
- Contract-first FastAPI API with Pydantic schemas and an OpenAPI export; the panels of a page share one fact sheet and their model calls run in parallel.
- Prompts as versioned YAML with a test that fails when a new prompt version is not pinned, and fail-open Langfuse tracing of scope, facts, narrative, model and prompt version for every summary.
- 98 offline pytest tests that run the real SQL against a synthetic fixture with a mock LLM.
- Tech stack: Python, FastAPI, Pydantic, SQLAlchemy, OpenAI, LangChain, Langfuse, Databricks SQL, pytest

Pharmacovigilance Risk Models: machine learning to predict the risks associated with late case reporting to Health Authorities.

- Designed and validated an XGBoost classification model to predict late case reporting, with SHAP analysis for explainability and root cause identification, clustering of the high- and low-risk buckets into patterns with decision-tree rules, and LLM-generated cluster descriptions. Deployed with Data Engineers in Databricks (DEV, QA and PROD), retrained every two months.
- Tech stack: Python, XGBoost, SHAP, Azure OpenAI, LangChain, Databricks, PySpark, Bitbucket

Aviation & Conversational AI

• **Client: Iberia Airlines (Spain) via Knowmad Mood (Spain)**

Development of **EVA (Enhanced Virtual Assistant)**, a multi-agent conversational ecosystem powered by LLMs to automate Iberia's customer service operations. Built four of its agents (Translator, Check-in, Tone, Flight Status) and contributed to the guardrail agent.

- LLM Engineering & Model Optimization: Built, tuned and deployed the Translator, Check-in, Tone and Flight Status agents on Azure OpenAI and OpenAI models (GPT-4.1, GPT-5.x) with AWS Bedrock (Nova Pro, Claude) as circuit-breaker fallback, and benchmarked candidate models on cost, latency and output quality.
- Multi-Agent Architecture: Built modular, hexagonal-architecture agents (ports and adapters, dependency injection) within an ecosystem of about fourteen agents (intent classification, guardrail, conversation manager, check-in, cancellation, flight status, flight documents, translator, tone, RAG) using LangChain and FastAPI with structured I/O through Pydantic schemas and DynamoDB conversation memory.
- API Integration: Integrated Iberia's check-in and flight-status APIs as LangChain structured tools, including an eleven-step check-in flow over eleven Iberia APIs, from reservation validation to boarding pass.

- Security & Guardrails: Tuned the prompt and multi-model configuration of the guardrail agent and built a red-team harness with prompt-injection, data-exfiltration, prompt-leakage and jailbreak datasets against the assistant platform.
- Evaluation & Monitoring: Load and regression testing of the assistant (Locust stress tests, a 1,000-prompt UAT battery), using the platform's per-call token, latency and cost tracking.
- Tech stack: Python, LangChain, Azure OpenAI, OpenAI, AWS Bedrock (Nova Pro, Claude), FastAPI, Pydantic, DynamoDB, Kinesis Firehose, structlog, pytest, Locust, GitLab CI/CD, ECS Fargate

Geolocation & Mobility Data Science

- **Client: The People Platform, now Numetrix (United States of America) via Truelogic Software (Uruguay)**
Numetrix (TPP) Text2SQL: production conversational analytics platform over US foot-traffic and mobility data, letting business users run their own analyses in natural language instead of queueing for a data analyst who can write SQL.
 - Pipeline Architecture: Four-stage hexagonal pipeline (query extraction, context-aware SQL generation, secure execution, natural-language transformation) with session memory and multi-turn follow-up resolution, plus FAISS semantic example selection for few-shot SQL generation.
 - Defence-in-Depth Guardrails: A regex input guard, a parallel LLM safety classifier that adds no perceptible latency, and a read-only SQL validator that blocks DML/DDL, including writes hidden in CTEs; validated with a 25-case adversarial battery (injection, jailbreak, exfiltration, obfuscation).
 - Quality & Auditability: A 301-case regression suite (301/301 passing) combining deterministic oracles with an LLM judge, an 18-case multi-turn follow-up corpus, pre-flight validation of extracted entities against the catalogs with fuzzy matching, and every turn persisted to a PostgreSQL audit table with per-stage timings.
 - Earlier work for the same client: a device home-location pipeline over tens of millions of devices a month (night-time location clustering, confidence scoring, demographic enrichment), first on AWS EMR and later on GCP Dataproc, Cloud SQL and BigQuery, and the first text2SQL agents (AWS Bedrock Titan, later OpenAI).
 - Tech stack: Python, LangChain, OpenAI, FAISS, LangSmith, Langfuse, PostgreSQL, BigQuery, GCS, Streamlit, Docker, Bitbucket Pipelines, PySpark, AWS (EMR, EC2, S3, RDS, Athena, Bedrock), GCP (Dataproc, Dataflow, Cloud Storage, Cloud SQL, Compute Engine)

Mining

- **Client: Codelco (Chile) via Option (Chile)**
 Development of a recommendation system to improve exploitation at the Chuquicamata Mine, the world's largest open pit copper mine in terms of excavated volume.
 - Analysed millions of tons of ore to discover trends and patterns, and built KPIs such as extraction uniformity and use of extraction points.
 - Built a mixed integer linear programming model (Pyomo) that recommends, per shift, the assignment of loaders to streets and extraction points to maximise adherence to the extraction plan.
 - Built and managed data engineering / data science pipelines with Kedro on Azure Databricks, with Azure DevOps CI building and publishing Docker images.
 - Tech stack: Python, Pyomo and GLPK, Kedro, Azure Databricks, PySpark, Delta Lake, MySQL, Azure DevOps, Docker, Git

Retail

- **Client: Falabella.com (Chile) via Option (Chile)**
 Modeling and transforming warehouse data to support recommendation systems for Falabella.com, the largest retailer in Latin America.
 - Analysed millions of sales transactions, identifying distinct customer clusters and purchasing patterns.
 - Built a collaborative-filtering recommendation system and automated the full pipeline from warehouse extraction to model deployment, with CI/CD.
 - Tech stack: Python, Git, GCP (Vertex AI, Cloud Storage, BigQuery, Cloud Functions, Cloud Composer).

Financial & Risk AI Solutions

- **Client: Pirani Risk (Colombia)**
 Consultant in the development of an Anti-Money Laundering (AML) product, through a two-phased method using PCA and Birch clustering to reduce feature complexity and identify groups within the banks' clients. Tech stack: Python, Git.
- **Client: Energage LLC, TopWorkPlaces.com (United States of America) via Oiga Technologies (Colombia)**

Consultant in a customer churn prediction project: identified the key features driving unsubscription and built the churn model that informed retention and personalisation decisions. Tech stack: Python, Snowflake, Git.

Transport & Logistics

- **Client: CFL Multimodal (Luxembourg) via University of Luxembourg**
Development, deployment and automation of ML models to predict disruptions in rail intermodal operations across several EU nations.
 - Built classification and regression models to predict delays in train operations, incorporating mixed integer programming to optimize solutions.
 - NLP Research: Led sentiment analysis projects on customer review data and published two research papers on tourist satisfaction towards transport and urban metro systems, applying LDA topic modelling and classification models.
 - Implemented MLOps to automate data extraction, transformation, model creation and improvement pipelines.
 - Tech stack: Python, optimization libraries, NLTK, Postgres, Docker, Apache Airflow, Git.
- **Clients: MetroValencia and Logitren (Spain) via Technical University of Valencia (Spain)**
Prediction and optimization of energy consumption in train operation for diesel-electric and electric trains: location-data analysis of Valencia metro and Valencia-Madrid freight services, ML models to predict consumption from driving practices, and optimization of operations to reduce it. Tech stack: Python, Matlab.
- **Clients: Government of Antioquia, Medellin City Hall and Antioquia Railway Department (Colombia)**
Geospatial data analysis and demand prediction for public infrastructure: railway route feasibility and prioritisation studies, geographical analysis of transport projects, and responsibility for the transport chapter of the Antioquia Development Plan 2020-2023. Tech stack: QGIS, Python.
- **Client: National University of Colombia (Colombia)**
Data scientist in several transport projects for public agencies in Medellin: analysis of demographic and survey data to understand passenger and freight travel behaviour, and classification models to predict traveller behaviour. Tech stack: R, Python, Matlab, Excel, Biogeme.

EDUCATION

2023	Postdoc – University of Luxembourg – Applied Machine Learning.
2019	Certification in Data Science, Correlation One – Data science for all (DS4A) program.
2017	PhD Eng., Technical University of Valencia, Spain – Applied Machine Learning [CUM LAUDE].
2013	M.Eng., National University of Colombia – Transportation.
2011	B.Eng., National University of Colombia – Civil engineering.

PUBLICATIONS

[Google Scholar](#)